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Tumor mutational burden and survival on immune checkpoint inhibition in >8000 patients across 24 cancer types

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ABSTRACT

Background There is uncertainty around clinical applicability of tumor mutational burden (TMB) across cancer types, in part because of inconsistency between TMB measurements from different platforms. The KEYNOTE 158 trial supported United States Food and Drug Administration (FDA) approval of the Foundation Medicine test (FoundationOneCDx) at TMB≥10 mut/Mb as a companion diagnostic (CDx) for single-agent pembrolizumab in second+line. Using a large real-world dataset with validated survival endpoint data, we evaluated clinical validity of TMB measurement by the test in over 8000 patients across 24 cancer types who received single-agent immune checkpoint inhibitor (ICI).

Methods Patients with advanced-stage cancers from 24 cancer types treated with single-agent anti-PD(L)1 therapy in standard-of-care settings were included. Deidentified data from electronic health records from approximately 280 cancer treatment facilities were captured into a clinico-genomic database. This study used the TMB algorithm from the FDA-approved test supporting solid tumor CDx and composite mortality variable validated against the national death index: real-world overall survival (rwOS). Following a prespecified analysis plan, rwOS by TMB level was assessed using Cox PH models adjusted for Eastern Cooperative Oncology Group performance status, prior treatment, microsatellite instability, sex, age, opioid rx pretherapy, and socioeconomic assessment.

Results 8440 patients met inclusion criteria. Adjusting for aforementioned factors, increasing TMB was significantly associated with rwOS across tumor types; HRs (95% CIs) relative to TMB<5: TMB 5 to <10: 0.95 (0.89 to 1.02), TMB 10 to <20: 0.79 (0.73 to 0.85), TMB≥20: 0.52 (0.47 to 0.58). For individual cancer types with prespecified statistical power, adjusted rwOS comparing TMB≥10 vs TMB<10 significantly favored TMB≥10 in 9 of 10 cancer types. In microsatellite stable subcohorts (except colorectal cancer), TMB≥10 remained associated with enriched ICI benefit. Exploratory assessments of patients receiving ICI+chemotherapy (n=4369) observed more favorable rwOS only in TMB≥20.

Conclusions Across >8000 patients treated with single-agent ICI, and within individual cancer types with sufficient power, elevated TMB based on the FDA-approved CDx was associated with more favorable rwOS compared with similar patients with lower TMB levels. This biomarker deserves further clinical investigation to potentially guide the use of immunotherapy in expanded clinical contexts.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Tumor mutational burden (TMB) has been investigated in many studies. However, considerable variability in assays and TMB algorithms, inconsistencies of prior study designs, and validity of clinical outcome measures have stymied generalizability of TMB clinical applicability.

WHAT THIS STUDY ADDS

⇒ Making use of a rigorous analysis plan and validated survival endpoint, this study adds validity to the use of the TMB measure that supports an FDA approval for a pan-tumor companion diagnostic, within its current context of use: treatment decisions for single-agent anti-PD-(L)1 axis immune checkpoint inhibitor (ICI). Associations were similar for second, third, and first-line use, but for assessments of anti-PD-(L)1 axis+chemotherapy, more favorable survival was observed only in higher TMB levels.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ TMB is an important biomarker correlative for next-generation immunotherapy development. Subgroup analyses of TMB-enriched populations in completed clinical trials might yield immediately useful insights for current clinical practice around ICI use.

INTRODUCTION

Immune checkpoint inhibitor therapies (ICIs) directed against PD-1/PD-L1 have revolutionized the therapeutic approach to many cancer types in a multitude of clinical settings. While immunohistochemistry-based PD-L1 expression scores have been integrated into regulatory approval as a

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predictive biomarker in certain cancers, the magnitude of ICI benefit differs considerably across different tumor types independent of biomarker selection. Further, there is significant variability in PD-L1 assays and interobserver variability, underscoring the need for orthogonal and/or composite biomarker development.[1–3]

In this regard, genomic predictive biomarkers do not face the same issues of interobserver repeatability and can be algorithmic. Drawing on initial discoveries of genomic associations with ICI outcomes[4 5] tumor mutational burden (TMB) was analytically and algorithmically refined for clinical use[6] and has been investigated as an alternative or add-on to PD-L1 assessment. The ICI pembrolizumab received pan-tumor United States Food and Drug Administration (FDA) approval for use in patients with metastatic solid cancers that have progressed following prior treatment, without satisfactory alternative treatment options, and with high TMB (≥10 mutations/Mb of sequenced DNA) using an FDA-approved TMB measurement.[7] The approval was primarily based on the KEYNOTE-158 study, which observed objective responses in 30 (29%; 95% CI 21% to 39%) of 102 patients in the TMB≥10 group and 43 (6%; 95% CI 5% to 8%) of 688 in the TMB<10 group[8] alongside a larger body of evidence supporting the clinical validity of TMB reviewed by the FDA as supplemental evidence.[9]

This approval has sparked some degree of controversy, with arguments emerging both in favor of and against: TMB cut-off of ≥10 not having a strong biological rationale and not being universally accepted in the literature; the approval was based on overall response rate, not overall survival, and outcomes data were limited for common disease areas such as breast, prostate, or colorectal carcinoma, and even more limited for some rarer cancer types.[10 11] Moreover, the clinical application of TMB assays is affected by DNA-based gene sequencing panel size, proper somatic and germline variant filtering, and other bioinformatic considerations,[12] as demonstrated by Friends of Cancer Research[13] and noted by a recent American Society of Clinical Oncology (ASCO) guidance.[14] These factors further hinder interstudy comparisons when using TMB measures generated by different laboratories and bioinformatic pipelines.

For these reasons, we sought to evaluate what is, to our knowledge, the largest cohort of patients to date that received single-agent anti-PD(L)1 inhibitors and had TMB measured by the same bioinformatic pipeline that supported the FDA approval of the TMB biomarker as a companion diagnostic. This study examines the associations of TMB with real-world overall survival (rwOS) to ICI monotherapy across and between different tumor types and explores clinical outcome associations with higher and lower levels of TMB.

METHODS

Study design and patient selection

The cohort comprised patients with advanced or metastatic cancers included in the nationwide US-based Flatiron Health—Foundation Medicine—deidentified clinico-genomic database (FH-FMI CGDB) between January 2011 and September 2022. All patients underwent genomic testing using FoundationOne, FoundationOneCDx, or FoundationOneHeme. The following chart-confirmed cancer types were included: anal carcinoma, breast cancer, cervix carcinoma, cholangiocarcinoma, colorectal cancer, cancer of unknown primary, endometrial cancer, gastroesophageal cancer, glioblastoma, hepatocellular carcinoma, head and neck squamous cell carcinoma, lung neuroendocrine carcinoma, melanoma, Merkel cell carcinoma, non-small cell lung carcinoma (NSCLC), ovarian/fallopian tube/primary peritoneal carcinoma, pancreatic adenocarcinoma, castration-resistant prostate cancer, renal cell carcinoma, soft tissue sarcoma, small cell lung carcinoma, skin carcinoma (unspecified), small intestine carcinoma, and bladder/urinary tract cancer. Deidentified clinical data originated from approximately 280 US cancer clinics (~800 sites of care, mostly community practice). Retrospective longitudinal clinical data were derived from electronic health records (EHRs), comprising patient-level structured and unstructured data, curated via technology-enabled abstraction of clinical notes and radiology/pathology reports, which were linked to genomic data derived from FMI comprehensive genomic profiling (CGP) tests in the FH-FMI CGDB by deidentified, deterministic matching.[15] Clinical data included demographics, clinical and laboratory features, timing of treatment exposure, treatment discontinuation, and date of death. The data are deidentified and subject to obligations to prevent reidentification and protect patient confidentiality.

To identify cancer types of interest, the Flatiron Health Pan-Tumor disease classification model was leveraged to harmonize chart-confirmed diagnoses as documented by the clinician into one of ~60 clinically meaningful solid-tumor cancer groups. This approach delineated cancers that are biologically and clinically distinct with an emphasis on the importance of histology in addition to tumor location. Inclusion and exclusion criteria were then applied: (1) a chart-confirmed diagnosis of metastatic disease (except for Glioma and other CNS tumors), (2) an FMI histology consistent with the chart-confirmed cancer type, (3) ICI monotherapy exposure and (4) a TMB test result performed on a tissue specimen. A cohort size cut-off of ≥10 patients was applied across cancer groupings such that 24 clinically distinct cancer types of interest were identified. Patients with more than one qualifying cancer type were included in the analysis based on the cancer type with the earliest diagnosis date.

Patient systemic antineoplastic lines of therapy were derived based on oncologist-defined heuristics. Patients were included in this study if they had a line of therapy comprising a single-agent anti-PD-1/PD-L1 axis

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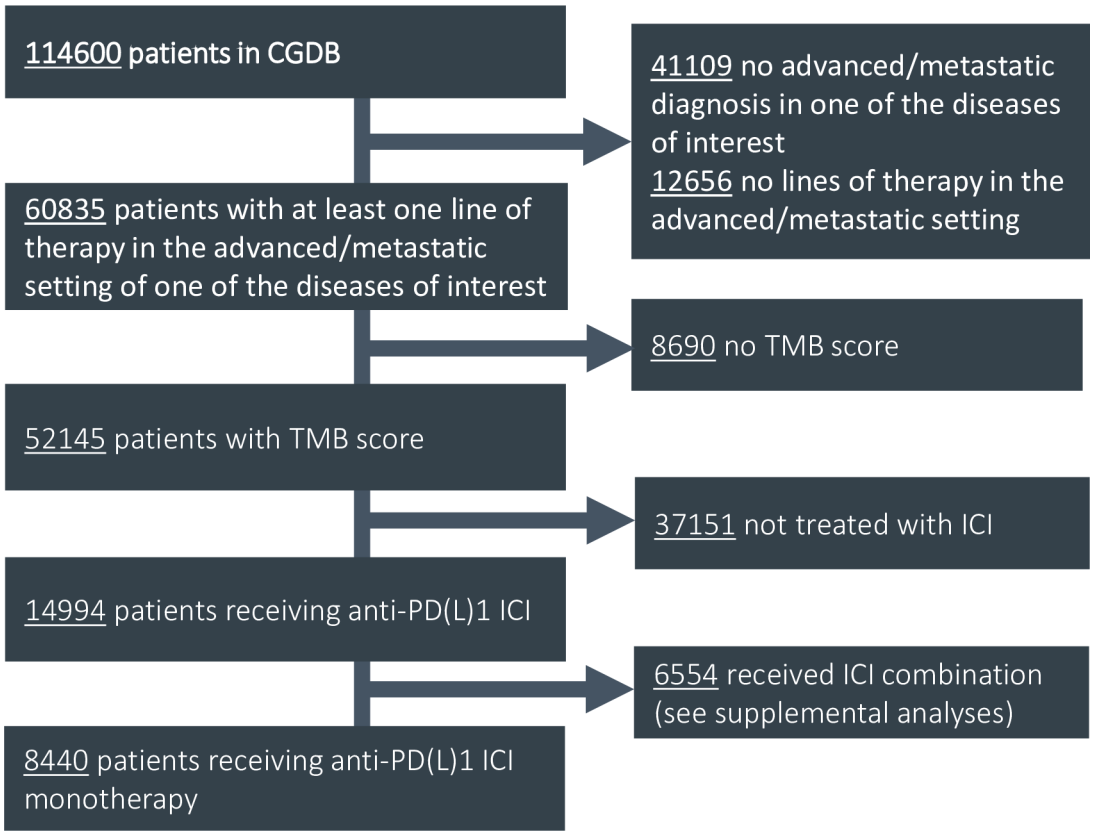

Figure 1 ICI monotherapy cohort overview. Cohort selection diagram. CGDB, clinico-genomic database; ICI, immune checkpoint inhibitor; TMB, tumor mutational burden.

monotherapy in the advanced or metastatic setting (but not maintenance setting) and had TMB assessed via tissue biopsy. Patients who received both ICI and chemotherapy in combination were evaluated in a separate exploratory analysis (figure 1). For patients who received anti-PD-1 axis therapy in multiple lines of therapy, only the earliest such line was used in this study.

Comprehensive genomic profiling

Hybrid capture-based next-generation sequencing (NGS) was performed on DNA extracted from patient tumor tissue specimens in a Clinical Laboratory Improvement Amendments–certified, College of American Pathologists-accredited laboratory (Foundation Medicine, Cambridge, Massachusetts, USA). Samples were evaluated for alterations as previously described.[16] TMB was determined on up to 1.1 Mb of sequenced DNA[6] with analytical validations reported in the Supplement of KEYNOTE-158[8]. Microsatellite instability (MSI) status by NGS was determined via an FDA-approved algorithm that calculates the fraction of unstable loci across >2000 loci.[17] Predominant genetic ancestry was determined by training a random forest classifier to distinguish the 5 ancestral superpopulations of the 1000 Genomes Project,[18] as previously described.[19]

Outcomes

rwOS was calculated from start of ICI to date of death from any cause, and patients with no record of mortality were right censored at the date of last clinic visit. Because patients cannot enter the database until a CGP report is delivered, rwOS risk intervals were risk set adjusted to the date of CGP report to account delayed cohort entry (immortal time).[20] Flatiron Health database date of death is a composite variable derived from three sources: documents within the EHR, Social Security Death Index, and a commercial death dataset mining data from obituaries and funeral homes, benchmarked to the National Death Index.[21] To preserve patient deidentification, death dates were available at month-detail, imputed to the 15th of the month. Real-world time to next treatment (rwTTNT) used in supplementary analyses was calculated from start of ICI treatment to initiation of next treatment or death if no additional treatment. Patients without treatment switch or death at time of last clinic visit date were censored at that date.

Statistical analysis

A prospectively declared statistical analysis plan was developed and executed. Consistent with International Society for Pharmacoeconomics and Outcomes Research guidelines,[22] the inclusion criteria, exclusion criteria, potential biases, primary outcome measures, exploratory outcome measures, handling of missing data, and all methods described below were prospectively specified prior to analysis execution unless otherwise noted. The statistical analysis

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plan included a pan-tumor assessment of outcomes associated with single-agent ICI in one multivariable model, and separately, per-tumor analysis.

Differences in time-to-event outcomes were assessed with the log-rank test and Cox proportional hazard (PH) models. While TMB is a continuous factor, it has highly skewed distributions that are challenging to normalize, and we, therefore, made use of categorical ranges of TMB for all Cox models. χ^2 tests and Wilcoxon rank sum tests were used to assess differences between groups of categorical and continuous variables, respectively. Multiple comparison adjustments were not performed; two-tailed p values are reported to quantify the strength of association for biomarker and each outcome, not for null hypothesis significance testing.

Missing values were incorporated as a separate categorical variable in all Cox models.

Associations with rwOS are sensitive to confounding by patient prognosis, and multiple steps were taken to adjust for potential imbalances in patient prognosis. Because cancers originating from different anatomical sites (ie, lung vs breast) can have dramatically different clinical course of disease, and the number of patients per cancer type is not equally distributed (which could bias estimates toward more common cancer types), our pan-tumor analysis made use of Cox models stratified by cancer type. In addition to adjustment by commonly used prognostic factors, we additionally included whether there is a record of the patient's provider prescribing an opioid within 2 months pretherapy, as a proxy for provider-intuited patient frailty not completely captured by other factors. Sex variable reflects a patient's sex at birth and does not reflect gender identity.

Pan-tumor analysis: Features included for adjustment in Cox models: Eastern Cooperative Oncology Group (ECOG) performance status (PS), line of therapy, sex, age, genetic ancestry, geographical area socioeconomic status (SES) assessment, opioid prescription pre-ICI therapy, and MSI status, stratified by cancer type.

Per-tumor analysis: Features included for adjustment in Cox models: ECOG PS, line of therapy, sex, age, genetic ancestry, geographical area SES assessment, opioid prescription pre-ICI therapy. Independence of TMB and MSI status was assessed via microsatellite stable (MSS)-only subgroup analyses. The analysis plan prespecified a threshold of at least 15 death events in both the TMB≥10 and TMB<10 groups to be present to be considered adequate power for assessment. In these analyses, we additionally chose to report an exploratory assessment of additional groups that had at minimum five death events in the TMB≥10 group. Those with less than 5 death events in the TMB≥10 group are not reported.

All analyses were conducted in R (V.4.2.1).

Comparative effectiveness summary

In order to contextualize our findings, a literature search was conducted for analyses fitting the following criteria: overall survival assessments comparing similar populations of patients who received ICI versus chemotherapy, with subgroup analyses reported stratified by TMB level.

RESULTS

Characteristics of analysis cohort

Out of 114 600 patient records in the database, 8440 were from patients receiving anti-PD-(L)1 ICI monotherapy in the advanced or metastatic setting for one of the cancers of interest and had available tissue-assessed TMB (figure 1). In addition, 6554 patients received ICI-containing combination therapy, of which 4369 received ICI+ chemotherapy combination therapy and were assessed separately as exploratory supplementary analyses. The overall cohort contains patients from 24 clinically distinct tumor types (table 1, online supplemental table 1) with a median age of 69 years and heterogeneous ECOG PS and geographical area-level SES across tumor types. Median follow-up was 31.7 months (95% CI 30.5 to 33.4 months).

Pan-tumor assessment of TMB association with clinical outcomes in ICI monotherapy cohort

Using a Cox PH model stratified by tumor type (see the "Methods" section), associations with rwOS were evaluated with respect to TMB ranges and other clinical factors in 8440 patients. Higher ECOG PS scores, greater number of prior lines of therapy, male sex, and prior opioid prescription pretherapy were all strongly associated with greater hazards of death (all p<0.001), with MSI-H status associated with reduced hazards of death (p<0.001) (figure 2). We observed that increasing TMB ranges were associated with stepwise decreasing hazards of death: compared with patients with TMB<5 mut/Mb, those with 5 to <10 had HR 0.95 (95% CI 0.89 to 1.02, p=0.153), those with 10 to <20 had HR 0.79 (95% CI 0.73 to 0.85, p<0.001), those with ≥20 had HR 0.52 (95% CI 0.47 to 0.58, p<0.001). In this model, adjusting for the aforementioned factors, age, genetic ancestry, and SES were not significantly associated with rwOS. Similar associations with outcome were observed when evaluating rwTTNT (online supplemental figure 1). Sensitivity analyses additionally evaluated subgroups of patients treated in either first line or second line or greater as separate models, and similar results were observed (online supplemental figure 2). Sensitivity analyses also observed similar associations of rwOS and TMB for PD-1 versus PD-L1 targeting drugs (online supplemental figure 3), as well as specific ICI agents (online supplemental figure 4). 1801 patients in the cohort additionally had PD-L1 data, and among this group of the cohort, both TMB and PD-L1 were significant contributors to a multivariable model additionally containing PD-L1 (online supplemental figure

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Table 1 ICI monotherapy cohort summary.

	TMB<10 (N=5411)	TMB 10+ (N=3029)	Total (N=8440)	P value
Tumor type				<0.001
NSCLC	2095 (38.7%)	1483 (49.0%)	3578 (42.4%)	
Anal	60 (1.1%)	16 (0.5%)	76 (0.9%)	
Breast	164 (3.0%)	64 (2.1%)	228 (2.7%)	
Cervical	72 (1.3%)	30 (1.0%)	102 (1.2%)	
Cholangio	56 (1.0%)	14 (0.5%)	70 (0.8%)	
crc	189 (3.5%)	214 (7.1%)	403 (4.8%)	
cup	62 (1.1%)	44 (1.5%)	106 (1.3%)	
Endometrial	110 (2.0%)	116 (3.8%)	226 (2.7%)	
Gastric	561 (10.4%)	91 (3.0%)	652 (7.7%)	
gbm	32 (0.6%)	11 (0.4%)	43 (0.5%)	
hcc	77 (1.4%)	1 (0.0%)	78 (0.9%)	
hnc	337 (6.2%)	59 (1.9%)	396 (4.7%)	
Inec	26 (0.5%)	16 (0.5%)	42 (0.5%)	
Melanoma	147 (2.7%)	381 (12.6%)	528 (6.3%)	
Merkel	33 (0.6%)	18 (0.6%)	51 (0.6%)	
Ovarian	112 (2.1%)	18 (0.6%)	130 (1.5%)	
Pancreatic	76 (1.4%)	14 (0.5%)	90 (1.1%)	
Prostate	50 (0.9%)	27 (0.9%)	77 (0.9%)	
rcc	296 (5.5%)	7 (0.2%)	303 (3.6%)	
Sarcoma	100 (1.8%)	14 (0.5%)	114 (1.4%)	
sclc	69 (1.3%)	39 (1.3%)	108 (1.3%)	
Skin	6 (0.1%)	8 (0.3%)	14 (0.2%)	
Smallint	14 (0.3%)	10 (0.3%)	24 (0.3%)	
Urothelial	667 (12.3%)	334 (11.0%)	1001 (11.9%)	
Age at Tx Initiation				<0.001
Median (Q1, Q3)	68.0 (60.0, 75.0)	69.0 (61.0, 76.0)	69.0 (61.0, 76.0)	
Practice type				0.017
Academic	659 (12.2%)	338 (11.2%)	997 (11.8%)	
Academic/community	68 (1.3%)	21 (0.7%)	89 (1.1%)	
Community	4684 (86.6%)	2670 (88.1%)	7354 (87.1%)	
EHR abstracted race				0.099
White	3791 (70.1%)	2129 (70.3%)	5920 (70.1%)	
Asian	123 (2.3%)	44 (1.5%)	167 (2.0%)	
Black or African American	315 (5.8%)	178 (5.9%)	493 (5.8%)	
Other race	844 (15.6%)	470 (15.5%)	1314 (15.6%)	
Unknown/not documented	338 (6.2%)	208 (6.9%)	546 (6.5%)	
Genetic ancestry				<0.001
AFR	397 (7.3%)	232 (7.7%)	629 (7.5%)	
AMR	327 (6.0%)	112 (3.7%)	439 (5.2%)	
EAS	144 (2.7%)	44 (1.5%)	188 (2.2%)	
EUR	4497 (83.1%)	2626 (86.7%)	7123 (84.4%)	
SAS	46 (0.9%)	15 (0.5%)	61 (0.7%)	
Socioeconomic status				0.164
1—Lowest SES	642 (11.9%)	383 (12.6%)	1025 (12.1%)	

Continued

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Table 1 Continued

	TMB<10 (N=5411)	TMB 10+ (N=3029)	Total (N=8440)	P value
2	798 (14.7%)	496 (16.4%)	1294 (15.3%)	
3	1005 (18.6%)	561 (18.5%)	1566 (18.6%)	
4	1065 (19.7%)	599 (19.8%)	1664 (19.7%)	
5—Highest SES	837 (15.5%)	443 (14.6%)	1280 (15.2%)	
Unknown	1064 (19.7%)	547 (18.1%)	1611 (19.1%)	
ECOG PS				0.162
0	1307 (24.2%)	788 (26.0%)	2095 (24.8%)	
1	2367 (43.7%)	1293 (42.7%)	3660 (43.4%)	
2+	1057 (19.5%)	600 (19.8%)	1657 (19.6%)	
Unknown	680 (12.6%)	348 (11.5%)	1028 (12.2%)	
Line of therapy				<0.001
First	1906 (35.2%)	1521 (50.2%)	3427 (40.6%)	
Second	1949 (36.0%)	1028 (33.9%)	2977 (35.3%)	
Third+	1556 (28.8%)	480 (15.8%)	2036 (24.1%)	
Sex				0.588
Female	2466 (45.6%)	1399 (46.2%)	3865 (45.8%)	
Male	2945 (54.4%)	1630 (53.8%)	4575 (54.2%)	
MSI by NGS				<0.001
MSS	5156 (95.3%)	2440 (80.6%)	7596 (90.0%)	
MSI-H	16 (0.3%)	459 (15.2%)	475 (5.6%)	
MSI-I	122 (2.3%)	77 (2.5%)	199 (2.4%)	
Unknown	117 (2.2%)	53 (1.7%)	170 (2.0%)	

Additional clinical factors are available in online supplemental table 1.
crc, colorectal cancer; cup, cancer of unknown primary; ECOG PS, Eastern Cooperative Oncology Group performance status; EHR, electronic health record; gbm, glioblastoma multiforme; hcc, hepatocellular carcinoma; hnc, head and neck cancer; lnec, large neuroendocrine carcinoma; MSI, microsatellite instability; MSS, microsatellite stable; NGS, next-generation sequencing; NSCLC, non-small cell lung cancer; rcc, renal cell carcinoma; sclc, small cell lung cancer; SES, socioeconomic status; smallint, small intestine .

5). Among patients with NSCLC, similar associations to the full cohort were seen for both those with squamous or non-squamous histology (online supplemental figure 6).

Per-tumor type assessment of TMB≥10 and survival on single-agent ICI

Individual Cox models were constructed per-tumor type, adjusting for ECOG PS, line of therapy, sex, age, and opioid prescription pretherapy, evaluating TMB<10 vs TMB≥10. Shown are the results for all patients (figure 3A) and the associated HR and 95% CI estimate per model. With the exception of SCLC (HR 0.89, 95% CI 0.44 to 1.09), all tumor types with adequate power for assessment (see the "Methods" section) including NSCLC, urothelial, melanoma, CRC, endometrial, gastric, breast, and carcinoma of unknown primary had point estimates and 95% CIs of less than 1, favoring TMB≥10. Assessing only the MSS patients in the cohort (figure 3B), all tumor types with adequate power maintained the same direction of effect, with CRC being a notable exception. Similar associations with outcome were observed when evaluating rwTTNT instead of rwOS (online supplemental figure 7).

Exploratory optimal TMB cutoffs

The distribution of TMB varies widely by tumor type (online supplemental figure 8). Cutoffs ≥10 and <10 were explored for 10 of the most common tumor types (figure 4). With the exception of melanoma, the point estimate was observed to decrease with increasing cut-off. Limiting the analysis to only patients also testing MSS, most tumor types continued to observe a decreasing point estimate with increasing cut-off (online supplemental figure 9).

Review of comparative effectiveness of ICI versus chemotherapy by TMB levels

The main focus of the statistical analysis plan was the assessment of the clinical validity of the associations between TMB and ICI monotherapy outcomes. A summary of previously reported analyses comparing outcomes of ICI versus chemotherapy is shown (figure 5). While the cohorts vary by tumor type, data type (eg, subgroup analysis from randomized controlled trials, comparative effectiveness of observational data) and specific ICI drugs, there exists a

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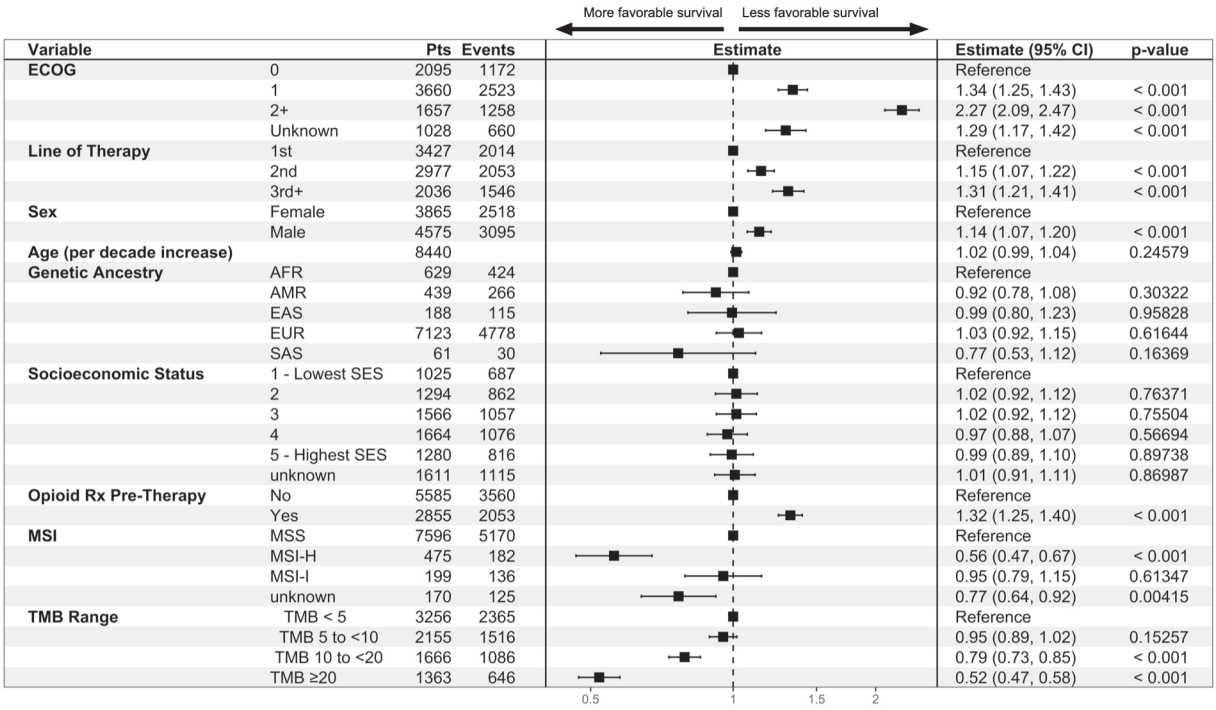

Figure 2 Pan-tumor assessment of the clinical validity of TMB to prognosticate rwOS on single-agent ICI. Stratified CoxPH models were used to assess the pan-tumor associations of observed features with rwOS. Plot shows HRs with 95% CI. Associations with time to next treatment are included in online supplemental figure 1). ECOG, Eastern Cooperative Oncology Group; ICI, immune checkpoint inhibitor; MSI, microsatellite instability; MSS, microsatellite stable; rwOS, real-world overall survival; SES, socioeconomic status; TMB, tumor mutational burden.

consistent pattern of reduced hazards of death on ICI monotherapy versus chemotherapy for patients with TMB≥10 (or equivalent whole exome sequencing), but this effect was not observed for those with low TMB.[9 23–30]

Per-tumor type assessment of TMB≥10 and survival on ICI and chemotherapy combination

In an exploratory analysis mirroring figure 1 (online supplemental figure 10–12, online supplemental table 2) among 4369 patients receiving ICI+chemotherapy combinations (out of 6554 patients treated with any ICI-containing combination), only the highest TMB ranges were associated with decreasing hazards of death: compared with patients with TMB<5, those with 5 to <10 had HR 1.04 (95% CI 0.94 to 1.14, p=0.436), those with 10 to <20 had HR 0.94 (95% CI 0.84 to 1.05, p=0.269), whereas those with ≥20 had HR 0.65 (95% CI 0.54 to 0.77, p<0.001).

DISCUSSION

Despite a multitude of reported studies, the value of TMB as a clinically applicable predictive biomarker for ICI therapy has been controversial. Prior work and meta-analyses evaluating TMB have reported inconsistent levels of clinical validity,[9 31–37] not surprising given the wide range of different TMB assays including differing amounts of DNA sequenced, algorithms for filtering germline variants, TMB cut points, clinical scenarios, and clinical outcome measures assessed. We report, to our knowledge, the largest biomarker association study to date of a real-world pan-tumor ICI-treated cohort making use of the TMB algorithm that has passed the analytical rigor of the FDA review process to support a companion diagnostic. With few possible exceptions, we observed that the effectiveness of ICI monotherapy increases with higher TMB levels across all tumor types, independent of MSI status; in MSS subcohorts (except colorectal cancer) TMB≥10 remained associated with enriched ICI benefit. Sensitivity analyses additionally suggest that the validity of TMB observed here is similar in first-line versus later-line settings (online supplemental figure 2) as well as potential for enrichment of outcomes in MSS colorectal cancer at higher TMB thresholds (online supplemental figure 9) seen in exploratory analyses.

Our assessment emphasized rwOS, thus differing from KEYNOTE-158 study, for which the primary endpoint was response rate. In contrast to our study, KEYNOTE-158 study did not demonstrate differences in OS, using a TMB cut point of TMB≥10, possibly related to its smaller sample size and cohort heterogeneity. The KEYNOTE-158 study, like ours, was a heterogeneous mixture of patients with many different cancer types and varying baseline prognostic characteristics that could have affected outcomes. Additionally, in KEYNOTE-158 study the TMB≥10 group was enriched for patients with small-cell lung cancer (33% for TMB≥10 vs 6% for TMB<10), which carries a particularly poor prognosis; this imbalance could bias the

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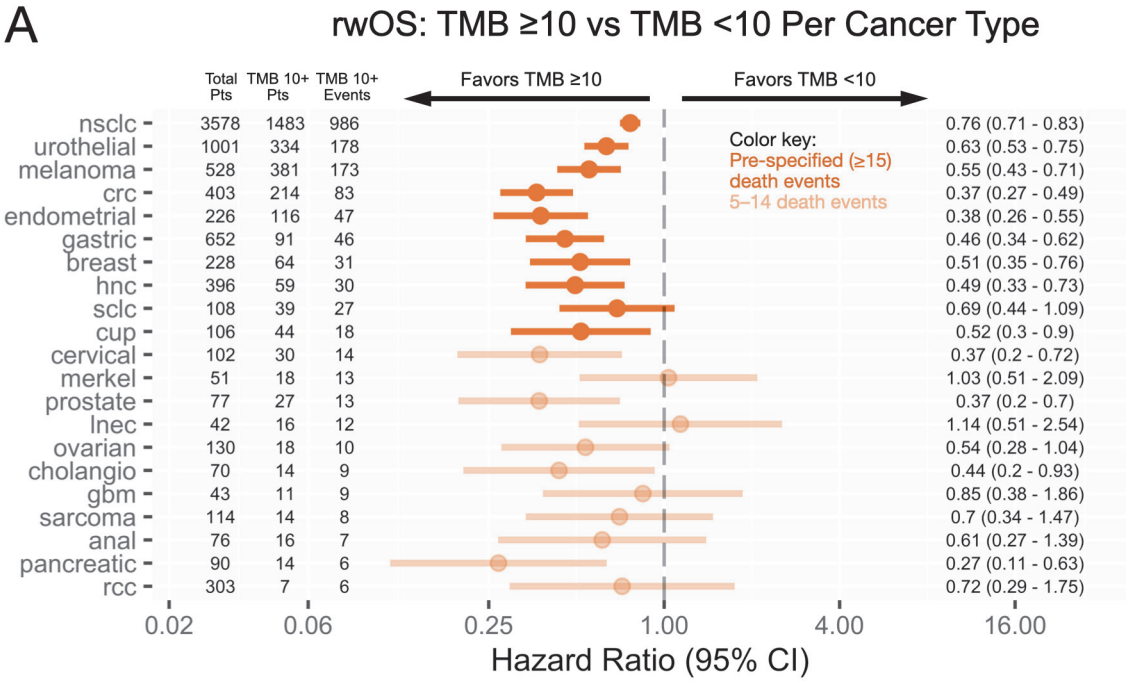

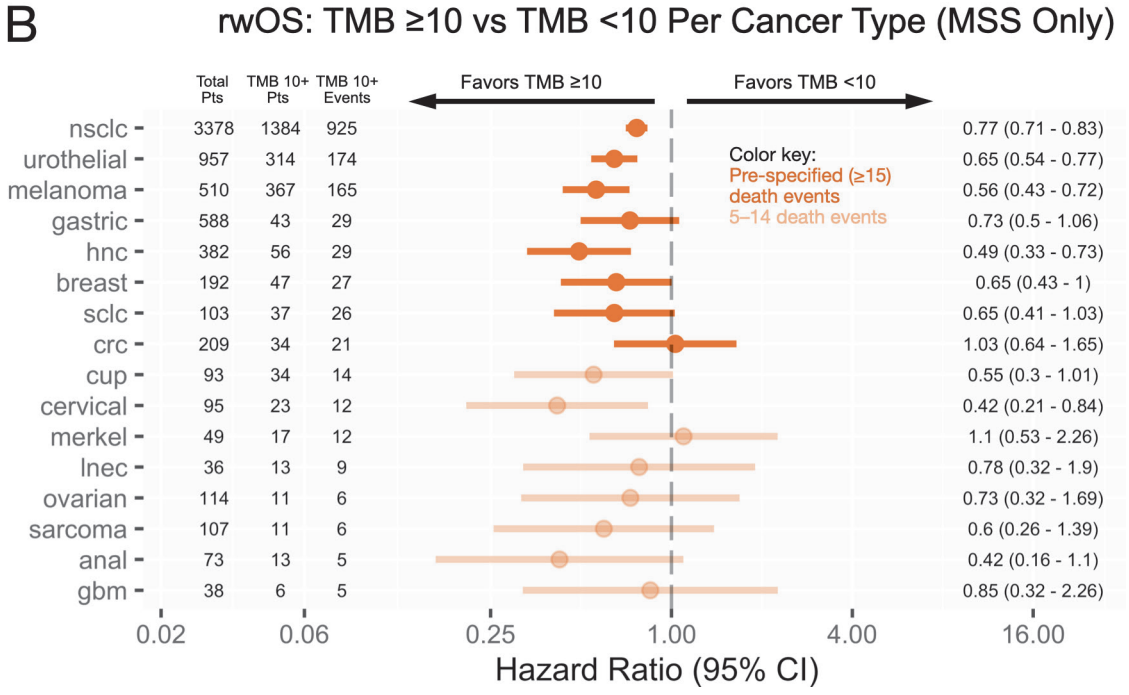

Figure 3 Per tumor type assessment of the clinical validity of TMB to prognosticate real-world overall survival on single-agent ICI. Cox models adjusted for ECOG PS, line of therapy, sex, age, and opioid prescription pretreatment are shown per disease type for rwOS for (A) all records meeting selection criteria and (B) the subgroup of these also testing MSS. The statistical analysis plan prespecified reporting any individual cancer type of at least 15 events. Any cancer type with at least five death events are shown, with forest plots colored by amount of power for assessment. Associations with real-world time to next treatment are included in online supplemental figure 3. crc, colorectal cancer; cup, cancer of unknown primary; ECOG PS, Eastern Cooperative Oncology Group performance status; gbm, glioblastoma multiforme; hcc, hepatocellular carcinoma; hnc, head and neck cancer; ICI, immune checkpoint inhibitor; lnec, large neuroendocrine carcinoma; MSS, microsatellite stable; nsclc, non-small cell lung cancer; rcc, renal cell carcinoma; rwOS, real-world overall survival.

PFS and OS analyses toward the null hypothesis as cancer type was not adjusted for in the PFS and OS analyses in KEYNOTE-158. In order to address these issues, our prospective statistical analysis plan made use of stratified Cox models that incorporated different baseline hazards of death per cancer type, as well as adjustment for standard

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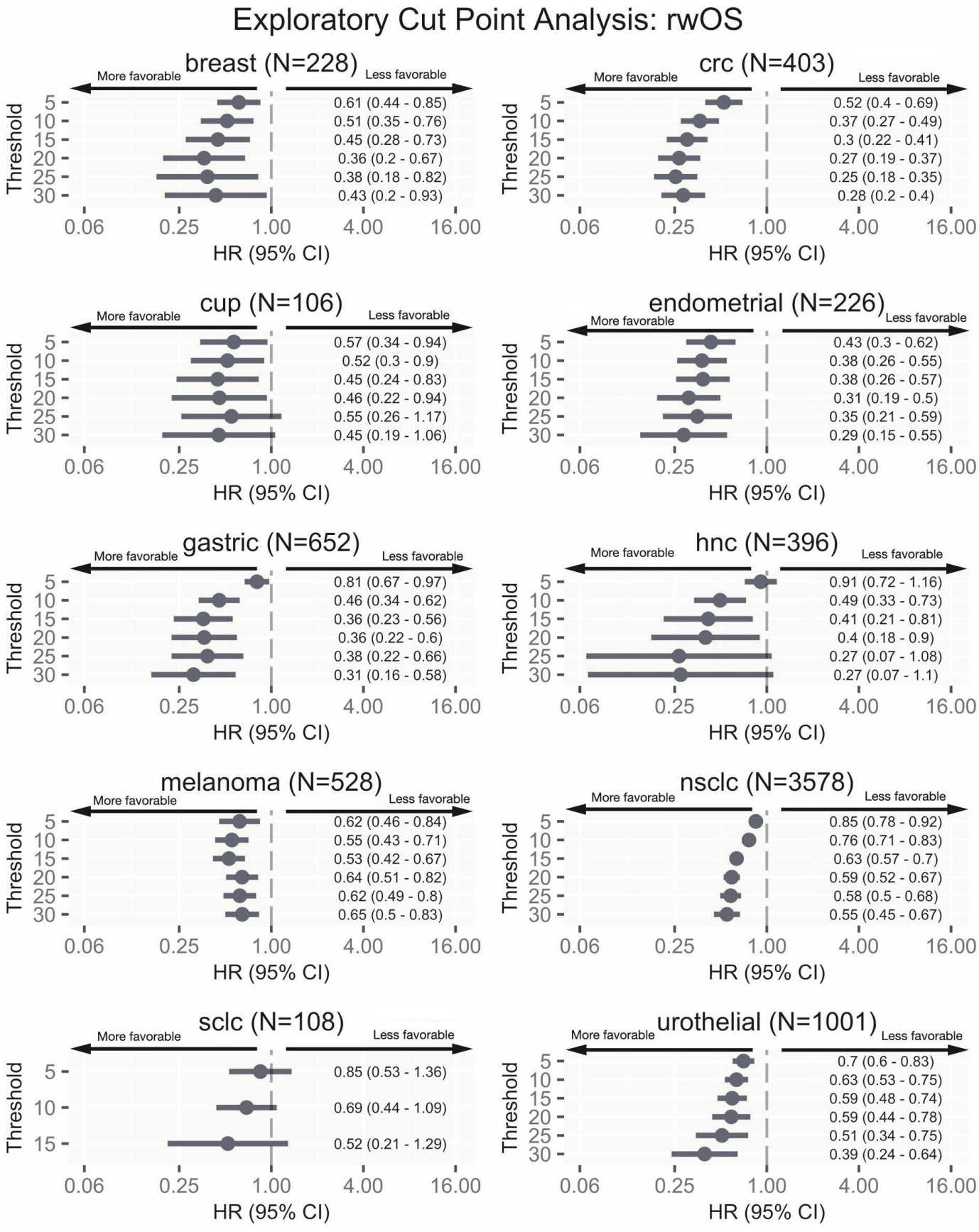

Figure 4 Exploratory cut-offs. Forest plots indicate the relative hazards of death when comparing patients grouped as "high" versus "low" TMB by the indicated threshold. Associations with patients testing MSS only are included in online supplemental figure 9. crc, colorectal cancer; cup, cancer of unknown primary; hnc, head and neck cancer; MSS, microsatellite stable; nsclc, non-small cell lung cancer; rwOS, real-world overall survival; sclc, small cell lung cancer; TMB, tumor mutational burden.

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Study	Tumor Type	Analysis Type	ICI	Comparator	TMB Assay	Threshold
Hill et al 2023 IJGC	endometrial (1L)	Comparative Efectiveness (CGDB)	Anti-PD(L)1	Platinum regimen	FMI	10 mut/mb
Lee et al 2022 CCR	gastric (1L)	RCT Subgroup Analysis (KN-062)	Anti-PD1	Platinum regimen	FMI	10 mut/mb
Lee et al 2022 CCR	gastric (1L)	RCT Subgroup Analysis (KN-062, no MSI-H)	Anti-PD1	Platinum regimen	FMI	10 mut/mb
Graf et al 2022 Cancer Res Comms	gastric (1L)	Comparative Efectiveness (CGDB)	Anti-PD(L)1	Platinum regimen	FMI	10 mut/mb
Shitara et al 2021 Annals of Oncology	gastric (2L)	RCT Subgroup Analysis (KN-061)	Anti-PD1	Paclitaxel	FMI	10 mut/mb
Shitara et al 2021 Annals of Oncology	gastric (2L)	RCT Subgroup Analysis (KN-061)	Anti-PD1	Paclitaxel	WES	175 mut/exome
Shitara et al 2021 Annals of Oncology	gastric (2L)	RCT Subgroup Analysis (KN-061, no MSI-H)	Anti-PD1	Paclitaxel	WES	175 mut/exome
Graf et al 2022 Cancer Res Comms	gastric (2L)	Comparative Efectiveness (CGDB)	Anti-PD(L)1	Non-platinum chemo	FMI	10 mut/mb
Herbst et al 2019 ESMO abstract	NSCLC (1L)	RCT Subgroup Analysis (KN-042)	Anti-PD1	Platinum regimen	WES	175 mut/exome
Marcus et al 2021 Clin Cancer Res	NSCLC (2L)	RCT Subgroup Analysis (KN-010)	Anti-PD1	Taxanes	WES	175 mut/exome
Graf et al 2022 JAMA Netw Open	prostate (2L+)	Comparative Efectiveness (CGDB)	Anti-PD(L)1	Taxanes	FMI	10 mut/mb
Winer et al 2020 ASCO abstract	TNBC breast (2L+)	RCT Subgroup Analysis (KN-119)	Anti-PD1	Non-platinum chemo	FMI	10 mut/mb
Galsky et al 2020 ASCO abstract	urothelial (1L)	RCT Subgroup Analysis (IMvigor130 interim OS)	Anti-PDL1	Platinum regimen	FMI	10 mut/mb
Wildsmith et al 2020 JITC abstract	urothelial (1L)	RCT Subgroup Analysis (DANUBE)	Anti-PDL1/CTLA4	Platinum regimen	FMI	10 mut/mb
Graf et al 2022 JCO-PO	urothelial (1L)	Comparative Efectiveness (CGDB)	Anti-PD(L)1	Carboplatin	FMI	10 mut/mb
Marcus et al 2021 Clin Cancer Res	urothelial (2L)	RCT Subgroup Analysis (KN-045)	Anti-PD1	Non-platinum chemo	WES	175 mut/exome

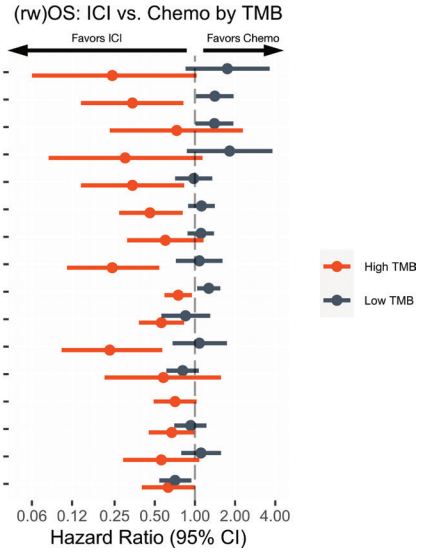

Figure 5 Summary of comparative analyses by TMB. A summary of published analyses comparing ICI versus chemotherapy by TMB levels. CGDB, clinico-genomic database; FMI, Foundation Medicine Incorporated; ICI, immune checkpoint inhibitor; KN, KEYNOTE; NSCLC, non-small cell lung cancer; RCT, randomized controlled trial; rwOS, real-world overall survival; TMB, tumor mutational burden; WES, whole exome sequencing.

prognostic factors (ECOG PS, prior treatment status) as well as additional features that might further capture physician-assessed patient frailty, such as a pretherapy opioid prescription. Complementing the overall analysis, our sample size was sufficient to report analyses per cancer type in a statistically rigorous manner, except for a small number of cancer types. The mortality measurement used in the rwOS analysis has also previously been benchmarked against the National Death Index[21] (see the "Methods" section), a level of validation that is not universal among real-world outcome data sources.

In addition to further evaluating the clinical validity of TMB as a predictive biomarker in multiple tumor types, a second main objective of this study was to explore the validity of the TMB cut-off at 10 mutations/Mb across a wide variety of tumor types. In this latter regard, our per-tumor type analysis used a single cut point (TMB≥10) for outcome association. Others have proposed that due to the wide variability in range of TMB values among different tumor types, that a binary assessment of TMB within each tumor type is a better paradigm for pan-tumor analysis of TMB. Using such an approach, Samstein *et al* reported a positive association of TMB by the Integrated Mutation Profiling of Actionable Cancer Targets assay (highest 20% in each tumor type) with overall survival for patients receiving ICI therapy, but not chemotherapy, across a wide range of tumor types,[33] concluding that there may not be a single universal definition of high TMB in a pan tumor context. Our large real-world analysis suggests that TMB≥10 is a reasonable single cut point for pan tumor assessment, when using the FDA-approved TMB measurement employed in the current study. However, the cut point of TMB≥10 was optimized to identify patients for whom have exhausted standard-of-care options and might be considered for immunotherapy (in line with the FDA label for pembrolizumab),[7] using the FDA-approved TMB measurement. The clinical decision at this point in care is typically either ICI or palliative care. Future potential use of TMB as a biomarker to decide between ICI or an active, established, efficacious therapy (such as first line chemotherapy) might require higher TMB levels to enrich for greater efficacy of ICI.

Interestingly, compared with TMB associations with outcome on single-agent ICI, we broadly observed a reduced magnitude of divergence between outcomes at lower TMB levels among those receiving ICI and chemotherapy combination, such that considerable enrichment for benefit occurred mostly in the TMB≥20 range (online supplemental figures 10–12). This observation can be further evaluated with subgroup analyses from existing, completed clinical trials, as few have yet evaluated the TMB cut point of 20.

A portion of patients in the cohort additionally had PD-L1 data, and a multivariable model containing TMB and PD-L1 resulted in both factors providing independent value to the model (online supplemental figure 5). This is consistent with prior studies suggesting the independent and additive clinical value of these two markers.[38 39]

Our study assessed survival associated with TMB and ICI use, but not how to use the biomarker or evaluate predictive nature in specific clinical contexts, such as comparing outcomes associated with ICI versus non-ICI by TMB level. While non-ICI outcomes data are available in the database, such assessments require rigorous clinical context-specific model building and careful considerations of unobserved confounders related to ICI versus non-ICI treatment assignments specific to narrow decision points in patient care.[22] Such studies have been published using the same database,[23 25 27 30] summarized in figure 5, alongside published subgroup analyses from completed

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randomized trials evaluating ICI monotherapy versus chemotherapy by TMB level. Future trials have the potential to enable immunotherapy-only or chemotherapy-free options for more patients. To this end, we provided an exploratory cut-point analysis to aid future prospective trial design and/or planned subgroup analyses (figure 4).

A recent ASCO Provisional Clinical Opinion on somatic genomic testing noted that discrepancies between laboratories may be identified, especially for complex mutational signatures.[14] There are ongoing efforts to harmonize and standardize TMB, such as those by the Friends of Cancer Research[13] that demonstrated a need for minimum panel size (>667 kb recommended) and proper pathogenic and germline variant filtering.[13] Corroborating these findings, a recent study by Nassar *et al* reported that a TMB algorithm with database-derived algorithmic germline filtering had reduced predictive ability in patients of non-European genomic ancestries, likely due to underrepresentation of these ancestries in publicly available databases.[32] In contrast, the FDA-approved TMB algorithm used in our study makes use of a genomic database for germline filtering that is much more representative of non-European ancestries compared with publicly available databases[19] used for germline filtering by TMB assays evaluated by Nassar *et al* and has been found by Huang, Graf, and Oxnard to have similar stratification of outcomes on ICI in both European and non-European ancestry.[31]

ICIs are among the most commonly used drugs in oncology. Biomarkers that can inform clinical decisions for ICI use have tremendous utility. While there are countless biomarkers in various stages of development, only a handful have passed the rigor of the FDA review process for routine clinical use.[40] Future versions of TMB might additionally incorporate HLA, more refined neoantigen load predictions, and mutational signatures.[41] Further, future innovation and biomarker development in the ICI space will likely be multimodal, including tumor, immune microenvironment, and host factors.[40] These innovations will need to be rigorously validated for routine clinical use, and we hope that our current study will provide a pan-tumor standard of comparison of performance.

Limitations of the current study

Observational and/or retrospective analyses are more prone to false discovery than prospective randomized trials, due to multiple hypothesis testing and potential imbalances between groups. We made use of a prospectively declared statistical analysis plan to prespecify hypotheses and analyses to reduce risk of false discovery and made rigorous adjustment of prognostic factors to reduce potential imbalances between groups that might confound survival comparisons. However, these adjustments do not account for all potential imbalances. We recognize that TMB calculation can vary considerably by panel size, gene content and bioinformatic filtering.[13] Therefore, the results from our study are not automatically applicable to other TMB algorithms.

CONCLUSIONS

Across >8000 patients treated with ICI, and within individual cancer types with sufficient power, elevated TMB based on an FDA-approved biomarker was associated with favorable rwOS compared with similar patients with lower TMB levels. Despite wide variability in the range of TMB levels among different tumor types, these real-world data support the clinical validity of TMB≥10 mut/Mb in patients receiving anti-PD-(L)1 monotherapy across multiple tumor types. Lastly, as ICI regimens in neoadjuvant or adjuvant settings become more common, the utility of TMB to help guide clinical decisions around ICI use deserves investigation to improve the precision of patient benefit.

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Correction notice This article has been corrected since it was first published. The Figure 4 'more favorable' and 'less favorable' labels were mistakenly switched and have now been corrected.

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have equity interest in Roche. Roche produces atezolizumab, an ICI. JSR also reports activity as a consultant and equity ownership in Tango Therapeutics and Celsius Therapeutics. JS, LB, CC-P and MP are employees of Flatiron Health, which is an independent member of the Roche Group, and stock ownership in Roche. JS also reports equity ownership in Flatiron Health. DG reports institutional research grants from Amgen, AstraZeneca, Genentech, Merck; and consultant/advisory board activity for Adagene (institutional), AstraZeneca (institutional), Roche-Genentech (institutional), Guardant Health (institutional), IO Biotech (institutional), Oncocyte (institutional), OncoHost (institutional), Merck (consultant). NA reports no conflicts since April 15, 2021. NA reports lifetime disclosures of consultancy to Astellas, Astra Zeneca, Aveo, Bayer, Bristol Myers Squibb, Calithera, Clovis, Eisai, Eli Lilly, EMD Serono, Exelixis, Foundation Medicine, Genentech, Gilead, Janssen, Merck, MEI Pharma, Nektar, Novartis, Pfizer, Pharmacyclics, and Seattle Genetics; and institutional research funding (lifetime) from Arnivas, Astellas, Astra Zeneca, Bavarian Nordic, Bayer, Bristol Myers Squibb, Calithera, Celldex, Clovis, Crispr, Eisai, Eli Lilly, EMD Serono, Exelixis, Genentech, Gilead, Glaxo Smith Kline, Immunomedics, Janssen, Lava, Medivation, Merck, Nektar, Neoleukin, New Link Genetics, Novartis, Oric, Pfizer, Prometheus, Rexahn, Roche, Sanofi, Seattle Genetics, Takeda, and Tracon.SG is a Consultant/Advisor for Bristol Myers Squibb, Bayer, Pfizer, EMD Sorono, Merck, Seattle Genetics, Gilead Sciences, Foundation Medicine, Guardant, Loxo Oncology. She is/has served on the Speakers Bureau for Gilead Sciences, Bristol Myers Squibb, Seattle Genetics and Janssen. She has stock options in Moderna, Buo Tech and Nektar Therapeutics. Her Institution has received research funding from Merck, Novartis, EMD Sorono, Gilead Sciences, Bristol Myers Squibb, Seattle Genetics, Moderna, Exelixis, Pfizer. SJK has served in a consultant/advisory role for Bristol Myers Squibb, Merck, Eli Lilly, Astellas, Daiichi-Sankyo, Pieris, Natera, Novartis, AstraZeneca, Mersana, Sanofi-Aventis, Servier, and Coherus. SJK reports stock/equity in Turning Point Therapeutics (ended 6/2022) and Nuvalent (ended 11/2022). MCA reports advisory board participation and honoraria from MSD Australia; honoraria from Pierre Fabre Australia and research funds to institution from MSD Australia and BMS Australia. AM is on speakers bureau with AstraZeneca and Exelixis. He has received honorarium from Taiho Oncology and Eisai and research funding from Pfizer. VS reports advisory board participation for Relay therapeutics, Roche, Pfizer, Bayer, Labcorp, Illumina, Jazz Pharmaceuticals, Aadi Biosciences. RE reports research funding (institutional) from Clovis Oncology, GSK, Merck, AstaZeneca; consulting fees (including participation on Data Safety Monitoring Board or Advisory Board) from Astra Zeneca, Clovis, Eisai, GSK, Immunogen, Mersana, Myriad, Novocure, Onconova, Elevar Therapeutics, Daichi Sankyo; payment for presentations/writing/educational events from Astra Zeneca, GSK, Immunogen, Merck, Medscape, CURIO, Great Debates and Updates; role as GOG P Associate Clinical Trial Advisor; role as Scientific and Medical Advisor to Clearity Foundation. DPC reports activity as a consultant for AstraZeneca, Nexus, Abbvie, Bayer, Boehringer Ingelheim, BMS, Daiichi Sankyo, EMD Serono, GSK, Incyte, Inivata, Inovio, Kyowa Hakko Kirin, Loxo, Merck, Novartis, Pfizer, Takeda, Flame, G1 Therapeutics, GenePlus, Genentech/Roche, Gloria Biosciences, Lilly, Oncocyte, OncoHost, Amgen, Curio Science, Johnson & Johnson, Sanofi, Novartis, Novocure, Janssen, Gritstone, Iovance. JWR reports consulting/advisory board activity for Blueprint, Beigene, Daiichi Sankyo, EMD Serano, Janssen, Regeneron, Sanofi, Biodesix, Bayer, Turning Point, Bristol Myers Squibb, Jazz Pharmaceuticals, Novartis, Roche/Genentech, Boehringer Ingelheim, Merck, SeaGen; and research funding (to institution) from Merck, Novartis, AstraZeneca, Spectrum, Revolution Medicines, Arrivent, IO Biotech, Vitrac. SS reports research funding from Astra Zeneca, Seagen, Eli Lilly, and Sermonix; and consulting for Eli Lilly, Astra Zeneca, Foundation Medicine, Gilead, Seagen, Pfizer, Novartis, Stemline, and Daiichi Sankyo.

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